

EXPERIENCE

SENTISUM | AI PRODUCT ENGINEER

August 2025 - Present (FTE) | February 2025 - July 2025 (Intern) | London, UK | Remote

- Built and scaled **askKYO** agent from **30 to 30,000+** tickets using clustering, intent detection, agent orchestration for highly-contextual responses.
- Automating topic exploration and classification pipelines to **eliminate manual work**, improving retrieval pipelines for enhanced answer generation.

UNIVERSITY OF BRITISH COLUMBIA | VISITING RESEARCHER

July 2024 - October 2024 | British Columbia, Canada | Internship

- Collaborated with Dr Fard and Dr Wu on **ClarifyCoder**, a framework enabling LLMs to ask clarifying questions during code generation.
- Developed novel data synthesis techniques to create ambiguous, incomplete, and inconsistent programming problems with clarifying Q&A; fine-tuned models to improve clarify-awareness, paper under review at ASE 2025.

MICROSOFT | TECHNICAL PROGRAM MANAGER

May 2024 - July 2024 | Hyderabad, India | Internship

- Led cross-functional agile teams to deliver **M365 Backup**. Planned future sprints, improved dashboard usability, defined KPIs, and introduced benchmarking. Drove roadmap planning, sprint execution, and aligned delivery with business goals and SDLC timeline.

APPLIED AI RESEARCH CENTRE, I-HUB | HEALTHCARE RESEARCHER

November 2023 - June 2024 | Hyderabad, India | Internship

- Built and managed image annotation tools using CVAT, hosted on Microsoft Azure. Applied OpenCV for image preprocessing and transformation to improve **detection of oral lesions** in mobile-captured images. Researched color spaces (RGB, LCV, HSL) to optimize annotation accuracy. Contributions supported a publication in **Scientific Reports (Nature) Journal**.

INFORMATION RETRIEVAL AND EXTRACTION LAB | RESEARCH ASSISTANT

May 2022 - July 2025 | IIIT Hyderabad, India

- Worked with Prof. Vasudeva Varma on **LLM evaluation**, machine-generated text detection, reference-free dialog metrics; co-authored **4 publications**.
- Mentored junior researchers through summer sessions and ongoing lab work, helping onboard them to research in NLP and language technologies.

PROJECTS

WHIZTAREEFA | FASTAPI, LLM-AGENT, RAG, GENAI, VECTORDB, SARVAM T2S API

- Developed an **RAG-enabled LLM-powered** QnA system, with agent actions, web search, and text-to-speech functionality, using ANN and fastAPI.

WIKISEARCH | TF-IDF, INFORMATION RETRIEVAL, LARGE SCALE DATA MANAGEMENT

- Built a **search engine on Wikipedia** pages, using parsed 90GB XML Wikipedia dumps, implementing inverted index creation and query-search mechanism.

TXTGENGUARD | ROBERTA, LONGFORMER, BILSTM, TML, PYTORCH LIGHTNING

- Built a mono & multilingual **LLM-generated text detector** using ensemble models (RoBERTa, DeBERTa, XLM-R) with traditional ML, achieving up to **85% accuracy**. Introduced predictability-based features (entropy, log-probs) from GPT2 variants, scaled training with PyTorch Lightning across 4 GPUs.

LOCKING-GRAPHS | SEMAPHORES, MUTEX, SYSTEM PROGRAMMING

- Utilizing Multi-Threading and parallelism to implement Greedy Graph Coloring Algorithm using coarse-grained & fine-grained locks, in C++.

EDUCATION

IIIT HYDERABAD

B.TECH. AND M.S. BY RESEARCH IN COMPUTER SCIENCE AND ENGINEERING (CGPA: 9.03/10)
Academic Awards: 1x Research List, 4x Dean's List
TA: Applications of Language Models (2025)
TA: Introduction to Human Sciences (2023, 2024)
TA: Behavioral Research Experimental Design (2023)
TA: Probability and Statistics (2022)
Host -- The IIIT Podcast S3

DELHI PUBLIC SCHOOL GHAZIABAD

CLASS XII: 97% | CLASS X: 10 CGPA | CBSE
Indo-German Student Exchange Program
Silver Plaque for Longstanding Academic Excellence (Scholar for 7 consecutive years)

SELECTED PUBLICATIONS

Understanding Robustness of LLM Evaluations under Perturbations

ICON 2024

Clarification-Aware Fine-Tuning for Programmatic Problem Solving

Under Review at TOSEM 2025

Screening of Oral Cancer Using Deep Learning Models

Scientific Reports (Nature) 2025

Leveraging Topical Embeddings for Annotation Detection in Tweets

SMM4H, ACL, 2024

ACHIEVEMENTS

Google Research Week: One of the top 70 research students selected to attend Google Research Week 2025, India.

Mitacs: Selected for the prestigious Globalink Research Internship, among the top 2% of applicants from India. Funded international research opportunity.

Silver Medal, AI for Social Good Hackathon, STASER Lab, UBC, Canada: Built CareBridge, a multilingual speech-enabled mental health assistant.

DRDOxIIIT: Invited to deliver a tutorial on modern language modeling and its defense applications to DRDO scientists.

OHAI: Organized 8th Oral History Conference, managed logistics, and delivered the concluding speech.

WikiMedia Summit: Organized and managed the Wikimedia Technology Summit, built website for the same.